



BIOFERTILIZERS — FULL MIND MAP

Topic-wise · Fully Explained · Hinglish · SSC / BPSC / BSSC / Railway

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1. Biofertilizers — Meaning & Main Examples

- Source biofertilizers ko living/latent cells of efficient microorganisms wali preparations batata hai jo crop plants ko nutrients uptake karne mein help karti hain. Azolla aur blue-green algae biofertilizers ke examples hain. [Q1]
- Rhizobium, blue-green algae (cyanobacteria) aur Azolla — teeno source ke mutabik biofertilizers hain aur soil nitrogen fixation se fertility improve karte hain. [Q3]
- Azolla biofertilizer hai; source Q6 mein Urea ko synthetic chemical fertilizer, 2,4-D ko herbicide aur Streptocycline ko agricultural antibiotic batata hai. [Q5, Q6]

EXAM TRAP — Q2 — Source Marks It Disputed

- ▶ Q2 mein source Azolla (1) + Blue-green algae (2) ko biofertilizers maanta hai; Alfalfa ko plant aur Nitrolin (calcium cyanamide) ko chemical fertilizer batata hai. “1 and 2 only” option na hone ki wajah se source question ko disputed/invalid mark karta hai. [Q2]

2. Nitrogen-Fixing Organisms — Nostoc, Anabaena & Azotobacter

- Nostoc, Anabaena aur Azotobacter nitrogen-fixing organisms hain. Mycorrhiza nitrogen-fixing organism nahi; source ise fungus–plant root symbiotic association batata hai jo mainly phosphorus aur water uptake mein help karta hai. [Q4]
- Blue-green algae ko Cyanobacteria bhi kaha jata hai aur source ise biological nitrogen fixation se associated batata hai. [Q9]
- Azotobacter free-living nitrogen-fixing bacterium aur biofertilizer hai — bioinsecticide nahi. Q8 mein isi wajah se “Azotobacter – Bioinsecticide” incorrect match hai. [Q8]

3. Blue-Green Algae / Cyanobacteria — Paddy Biofertilizer

- Cyanobacteria / blue-green algae atmospheric nitrogen fix karte hain aur water-logged paddy (rice) fields mein chiefly biofertilizer ke roop mein use hote hain. [Q10, Q11]
- Source Q14 mein Cyanobacteria ko microorganisms used as biofertilizers ka correct option batata hai, especially rice cultivation mein nitrogen-fixing capability ki wajah se. [Q14]

4. Azolla — Aquatic Fern, Anabaena Association & Uses

- Azolla ek small, free-floating aquatic fern hai. Source ise paddy cultivation mein widely used nitrogen-rich biofertilizer batata hai. [Q5, Q16, Q17, Q18]
- Azolla ka nitrogen-fixing benefit symbiotic blue-green alga / cyanobacterium Anabaena azollae se associated hai; source Azolla + blue-green algae association ko good biofertilizer batata hai. [Q13, Q15]

- Source Azolla ko livestock ke liye supplementary food, especially milking bovines ke liye, high-protein feed ke roop mein bhi describe karta hai. [Q7]

5. Crop–Biofertilizer Matching

- Source ke correct crop pairs: Blue-green algae → Paddy; Rhizobium leguminosarum → Pea; Azotobacter → Wheat. Azolla → Maize incorrect pair hai, kyunki source Azolla ko mainly paddy se associate karta hai. [Q12]

⚠ EXAM TRAP — Paddy Associations

- ▶ Blue-green algae / Cyanobacteria → Paddy; Azolla → Paddy. Source Q12 mein Azolla–Maize ko incorrect match batata hai. [Q10, Q11, Q12, Q18]

6. VAM / Mycorrhiza — Phosphatic Biofertilizer

- Vascular Arbuscular Mycorrhiza (VAM) phosphorus uptake improve karta hai, isliye source ise phosphatic biofertilizer batata hai. VAM fungi water aur poorly mobile ions, especially phosphorus, ki uptake mein host plant ko benefit dete hain. [Q19]

7. Biochar — Farming Uses

- Biochar source ke mutabik stable, carbon-rich charcoal hai jo high-temperature, low-oxygen pyrolysis se banta hai. Ye vertical farming ke growing medium ka part ho sakta hai, nitrogen-fixing microorganisms ko promote kar sakta hai aur water/water-soluble nutrients ko longer retain karne mein help karta hai. [Q20]

8. Compost, Earthworms & Vermicompost

- Compost mein biodegradable farm waste pits mein decompose hota hai aur organic matter/nutrients se rich hota hai. Source ke mutabik earthworms aur microbes decomposition ko slow nahi, accelerate karte hain. [Q21]
- Earthworms ko source “Ploughers of the soil” / farmer’s friends batata hai: burrowing se soil aeration aur air-water movement improve hota hai, aur organic matter ko nutrient-rich vermicompost mein break down karne mein help karte hain. [Q28]

9. Nitrogen Use Efficiency in Crop Production

- Source ke mutabik nitrogen-use efficiency split applications, nitrogen inhibitors aur slow-release nitrogen fertilizers — teeno se improve ki ja sakti hai. [Q22]

10. Legumes, Rhizobium & Nitrogen-Fixing Plants

- Nitrogen-fixing bacteria jaise Rhizobium legumes/pulses ki roots mein symbiotically root nodules banakar rehte hain. [Q23]
- Source ke listed nitrogen-fixing plants mein Alfalfa, Chickpea aur Clover hain; Amaranth, Purslane (Kulfa) aur Spinach ko nitrogen-fixing plants nahi bataya gaya. [Q24]
- Gram, pea aur bean jaise legume crops nitrogen fixation mein helpful hain; source paddy plant ko itself nitrogen-fixing nahi maanta. [Q25]
- Soyabean mein nitrogen fixation ke liye source Rhizobium japonicum (Bradyrhizobium japonicum) ko responsible batata hai. [Q26]

11. Leg-haemoglobin in Root Nodules

- Leg-haemoglobin leguminous plants ke nitrogen-fixing root nodules mein milta hai. Source ise oxygen-carrying phytohemoglobin batata hai jo nodule bacteria ko sufficient oxygen regulate/supply karta hai while oxygen-sensitive nitrogenase ko protect karne mein help karta hai. [Q27]